DP Biology – D3.1 Plant Systems

Reproduction in plants

- What does sexual or asexual reproduction show about change and continuity?
- How must organisms change in order to reproduce?

Plant statements D3.1.8–12, SL and HL. Textbook pp. 702–708.
Image: honey bee on a flower. Virtual gamma, CC BY-SA 3.0.

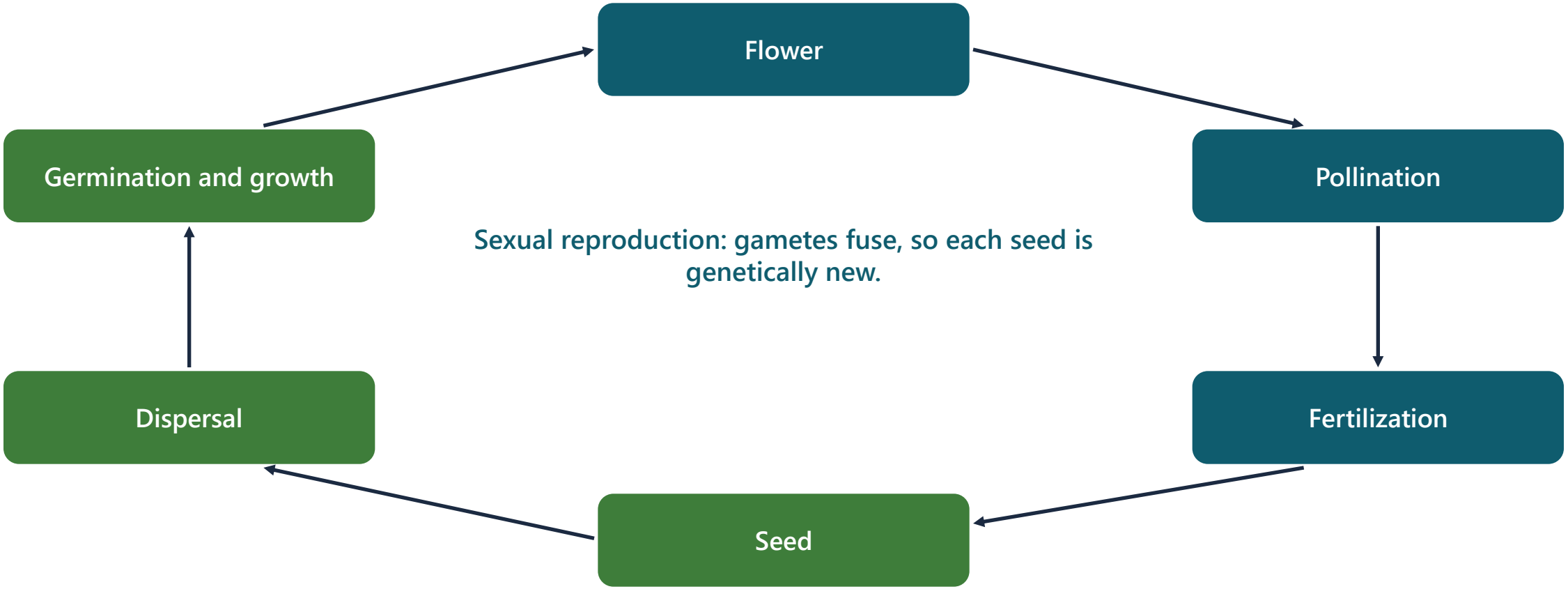


Objectives for each class

Class	Statements	By the end of the class, you can...
1	D3.1.8–9	Outline sexual reproduction in flowering plants, and draw an annotated insect-pollinated flower
2	D3.1.10–12	Explain how cross-pollination is promoted, and how seeds disperse and germinate

The life of a flowering plant

D3.1.8

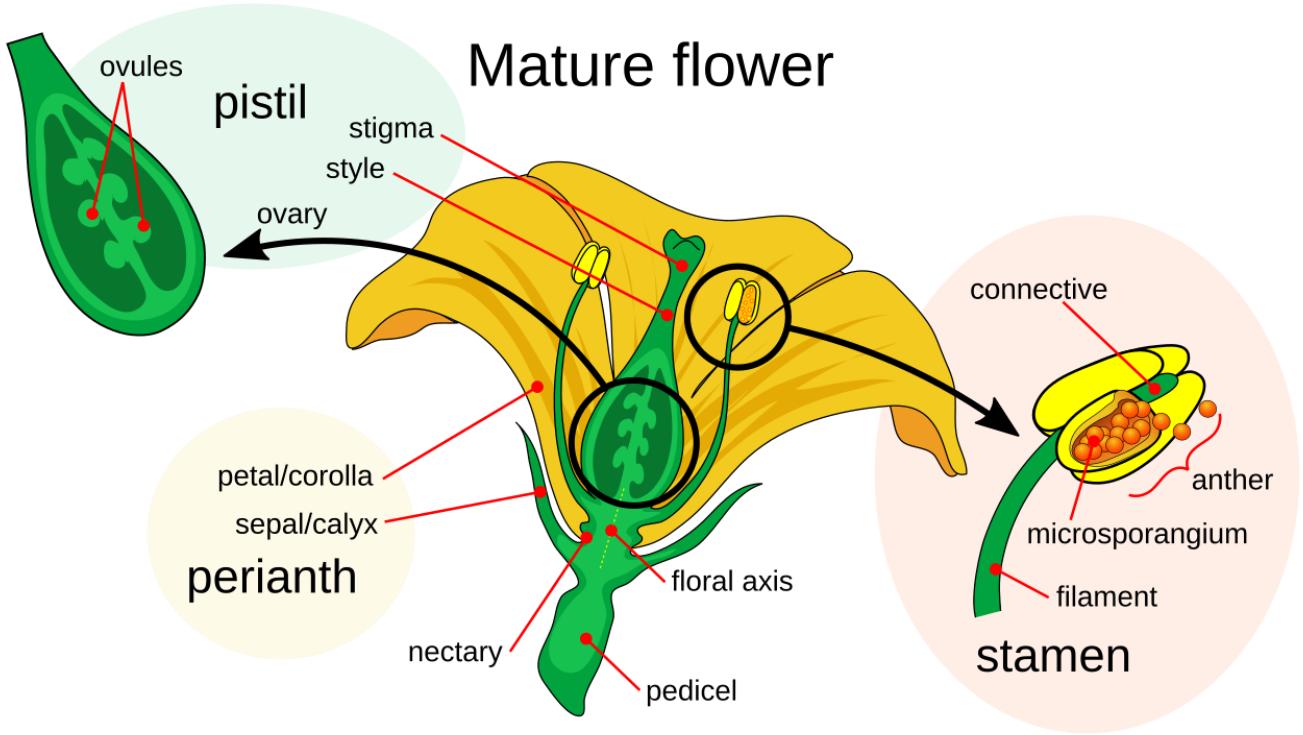


Key idea

Pollination moves pollen; fertilization fuses gametes; dispersal moves seeds. Keep the three apart.

An insect-pollinated flower

D3.1.9



A mature flower. Mariana Ruiz, LadyofHats, public domain.

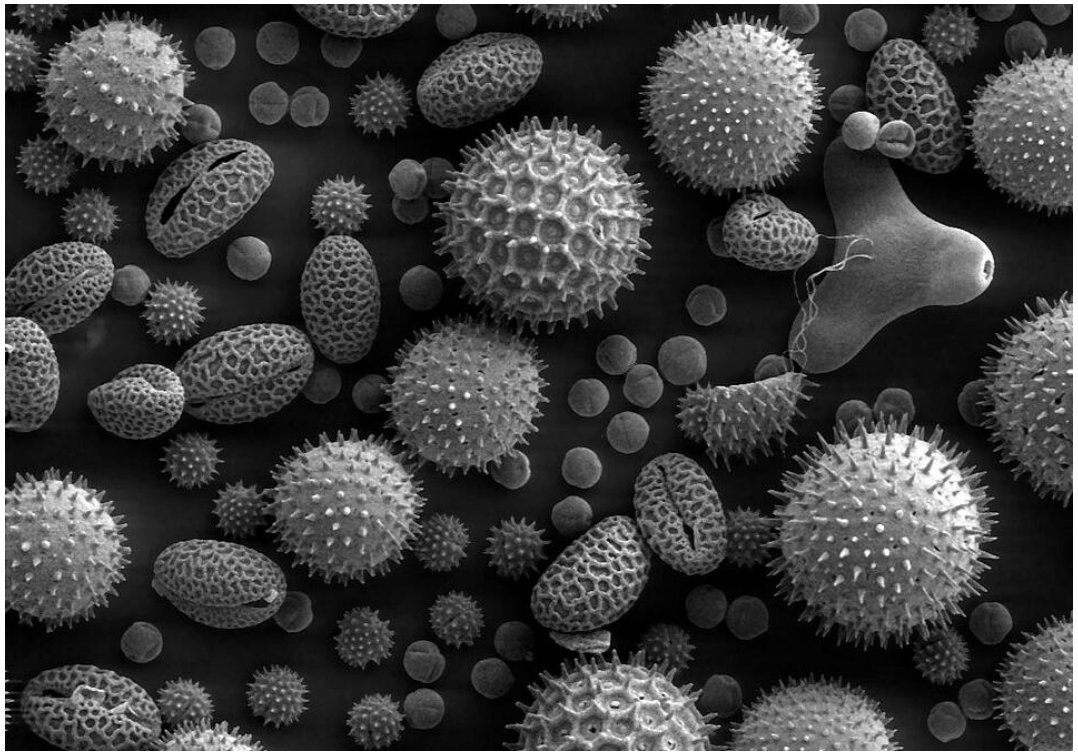
Part	Function
Sepal	Protects the bud
Petal	Attracts pollinators
Anther	Makes pollen
Filament	Holds up the anther
Stigma	Pollen lands here
Style	Holds up the stigma
Ovary	Contains ovules
Nectary	Nectar as a reward

Exam note

Stamen = anther + filament. Carpel = stigma + style + ovary. Be ready to draw and annotate.

Where the gametes are made

D3.1.8



Pollen grains, coloured electron micrograph. Dartmouth College, public domain.

Many flowers are **hermaphrodite**: male and female parts in one flower.

Male gametes form inside **pollen grains** in the anthers; the egg forms inside an **ovule** in the ovary. Meiosis comes first, so gametes are haploid.

Key idea

Even self-fertilization is sexual reproduction, because gametes fuse.

From pollination to embryo

D3.1.8

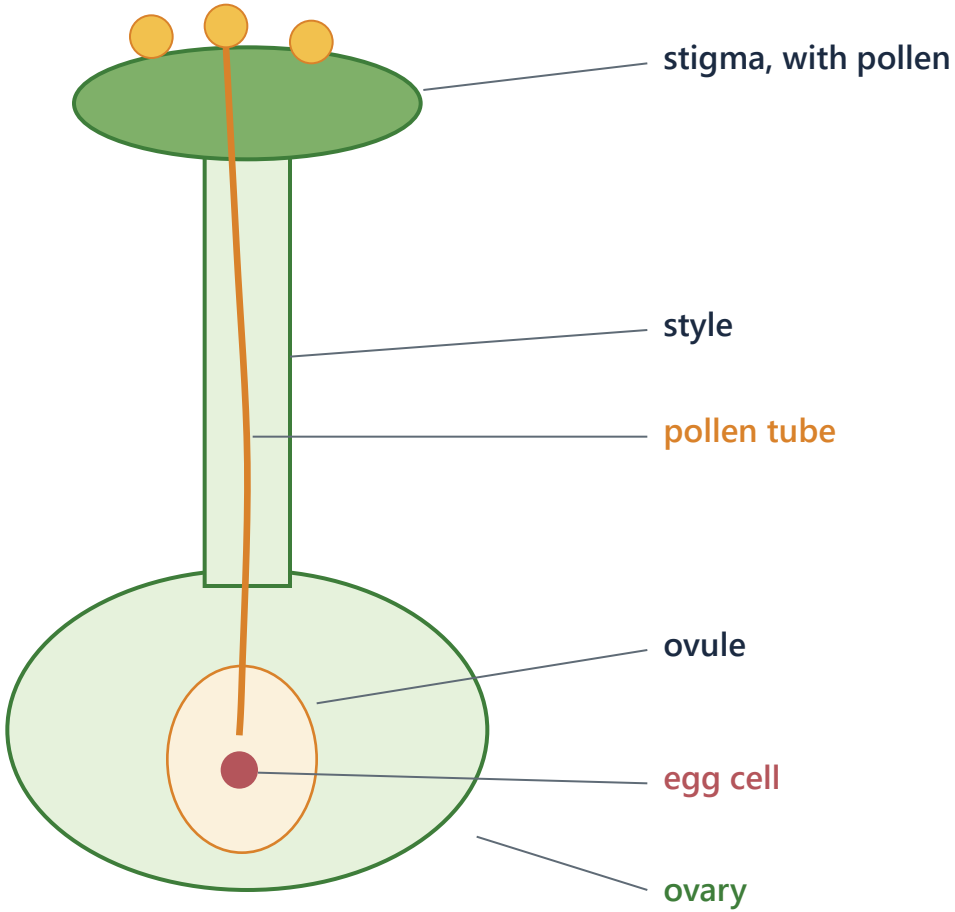


The ovule then becomes the **seed** and the ovary becomes the **fruit**.

Common mistake

Pollination delivers pollen to the stigma, not the ovary. The pollen tube carries the male nuclei down to the ovule.

The pollen tube



Drawn diagram of a carpel, not to scale.

A pollen grain on the stigma grows a **tube** down through the style. It carries male nuclei to the ovule.

One male nucleus fuses with the egg: **fertilization**.

Adapted for insect pollination

D3.1.9



Honey bee on a flower. Virtual gamma, CC BY-SA 3.0.

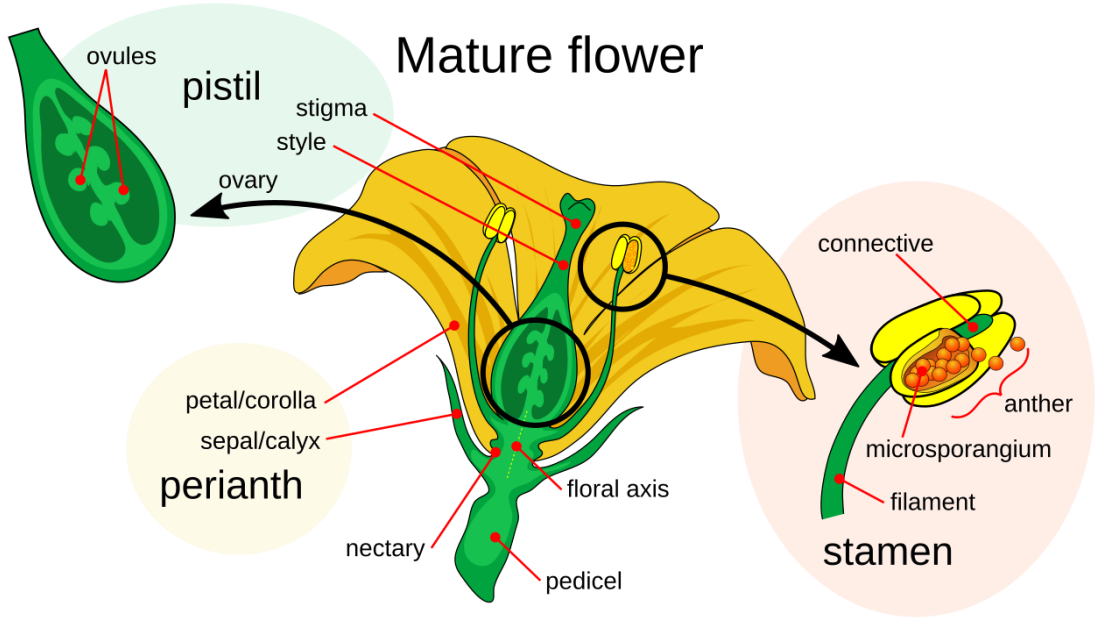
Feature	How it helps
Bright petals, scent	Attract insects
Nectar	Reward that brings insects back
Sticky pollen	Clings to the insect's body
Anthers and stigma inside	Insect brushes past both

Mutualism

The insect gets food; the plant gets its pollen carried to another flower of the same species.

Practice: name the part

D3.1.9



A mature flower. Mariana Ruiz, LadyofHats, public domain.

1. Where is pollen made?
2. Which part receives pollen?
3. Through which part does the pollen tube grow?
4. Which part becomes the seed?
5. Which part becomes the fruit?
6. Which part rewards visiting insects?

Check-in: class 1

D3.1.8–9

1. Distinguish between pollination and fertilization.
2. State where male and female gametes are made in a flower.
3. Describe the role of the pollen tube.
4. Describe three adaptations of a flower that relies on insects to carry its pollen.
5. State what the ovule and the ovary become after fertilization.

How to use this

Answer without your notes, then check against slides 3–9.

Self-pollination and cross-pollination

D3.1.10

	Self-pollination	Cross-pollination
Pollen comes from	The same plant	Another plant, same species
Result	Inbreeding	Outbreeding
Advantage	Keeps a genome suited to a stable place	More variation: can adapt
Disadvantage	Less variation; harmful recessives meet	Pollen must reach a stigma by chance

Key idea

Animals or wind carry pollen between plants. Most plants promote cross-pollination.

How plants promote cross-pollination

D3.1.10

Method	Example
Anthers and stigma ripen at different times	Pecan trees
Stigma held above the anthers	Many flowers
Separate male and female flowers	Birch trees
Separate male and female plants	Holly

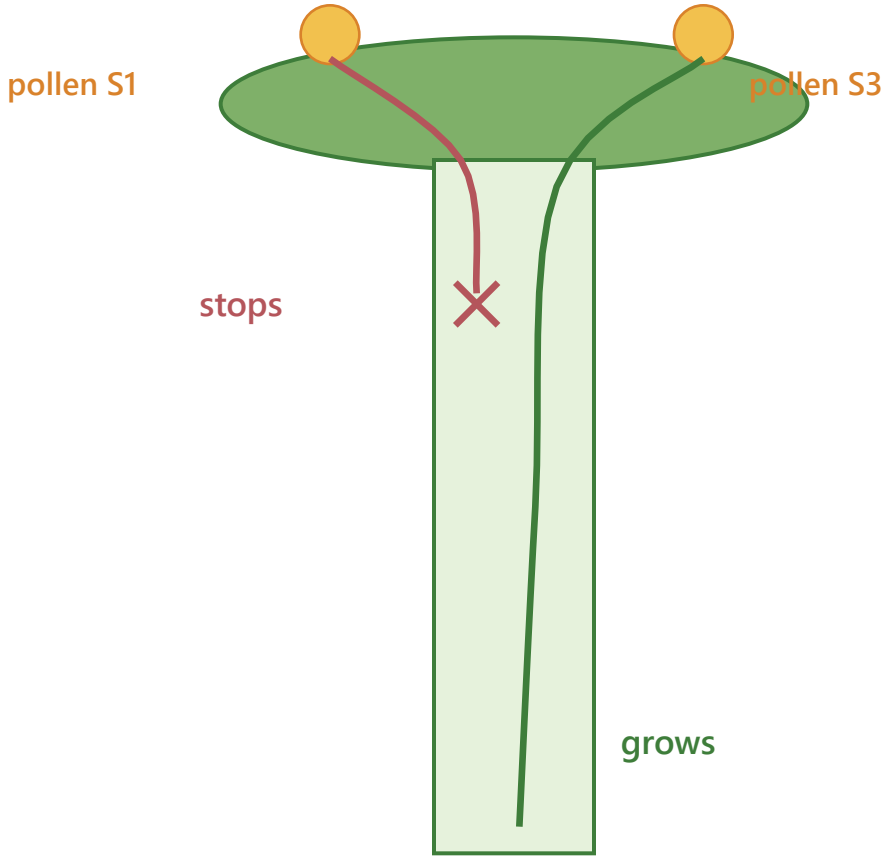


Holly. James E. Petts, CC BY-SA 2.0.

Think

A gardener plants one holly bush and it never has berries. Explain why, and suggest a fix.

Self-incompatibility



Style of a plant with alleles S1 S2

Many species **reject their own pollen**. If a pollen grain carries an S allele also present in the style, its tube stops growing.

So the gametes that fuse come from **different plants**: more variation, less inbreeding.

Style	Pollen allele	Tube
S1 S2	S1 or S2	Stops
S1 S2	S3	Grows

Seed dispersal

D3.1.12





Dandelion seeds, carried by wind (CC BY-SA 3.0); coconut, carried by water (Kulmalukko, CC BY 3.0).

Method	Example
Wind	Dandelion: parachute of hairs
Water	Coconut: floats
Animal	Fleshy fruit eaten
Explosive	Pod bursts as it dries

Common mistake

Pollination moves pollen before fertilization. Dispersal moves seeds after it, away from the parent, so seedlings compete less for light, water and nutrients.

Inside a seed

D3.1.12



Germinating beans. Doronenko, CC BY-SA 3.0.

Part	Function
Testa	Protective seed coat
Cotyledons	Store food for the embryo
Embryo root	Becomes the first root
Embryo shoot	Becomes stem and leaves
Micropyle	Gap where water enters

Key idea

A dry seed is dormant: only 10–15% water, so metabolism is very slow.

What germination needs

D3.1.12

Water

Rehydrates cells; the embryo becomes active

Oxygen

For aerobic respiration, which supplies ATP

Warmth

Near the optimum for the enzymes



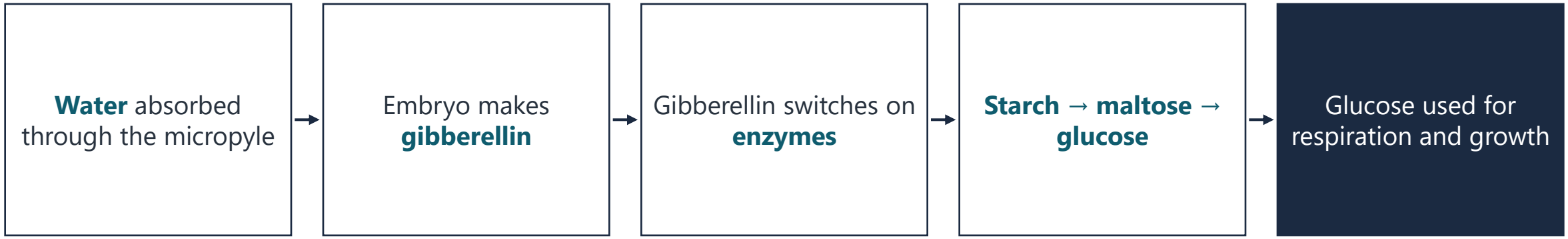
Seedlings. Arn, CC BY-SA 4.0.

Activity

Design an experiment to test one condition. How many seeds per dish? What will you control? How will you allow for seeds that are dead or dormant?

Mobilizing the food reserves

D3.1.12



Proteins are broken down to amino acids too. The seedling lives on these reserves until its leaves can photosynthesize.

Exam note

A germinating seed loses dry mass at first: respiration of stored food releases carbon dioxide before photosynthesis begins.

Working with data: temperature and germination

D3.1.12

Percentage of bean seeds germinated after 7 days. These data are invented for practice; the pattern matches real results.

Temperature / °C	Germinated / %
5	4
15	52
25	94
35	70
45	0

1. Identify the optimum temperature.
2. Explain the results at 5 °C and at 45 °C.
3. Suggest a control variable and how to keep it constant.

Practice: dry mass of seedlings

D3.1.12

Mean dry mass of bean seedlings grown in the light. The values are invented for practice; the pattern is realistic.

Day	Dry mass / g
0	2.0
3	1.8
6	1.7
9	1.9
12	2.3

1. Describe the change in dry mass.
2. Explain the fall from day 0 to day 6.
3. Explain the rise after day 6.
4. Predict the result for seedlings kept in the dark.

Check-in: class 2

D3.1.10–12

1. Explain why cross-pollination is usually an advantage.
2. Give two ways plants promote cross-pollination.
3. Explain how self-incompatibility prevents self-fertilization.
4. Distinguish between pollination and seed dispersal.
5. List the conditions needed for germination, and why each is needed.

How to use this

Answer without your notes, then check against slides 11–19.

Common mistakes to avoid

Not this	But this
Pollination and fertilization are the same.	Pollination moves pollen; fertilization fuses gametes.
Pollen is the male gamete.	Pollen grains carry the male gametes.
Seeds need light to germinate.	Most need water, oxygen and warmth.
Self-pollination gives more variation.	Cross-pollination gives more variation.

Summary: reproduction in plants

D3.1.8–12

	Key idea
D3.1.8	Pollination, pollen tube, fertilization, embryo; ovule → seed
D3.1.9	Petals, nectar, sticky pollen: adapted to insects
D3.1.10	Cross-pollination by timing, position or separate sexes
D3.1.11	Pollen with a matching S allele is rejected
D3.1.12	Dispersal; germination needs water, oxygen, warmth

Check that you can...

1. **Draw** and **label** an insect-pollinated flower
2. **Outline** the events from pollination to the formation of a seed
3. **Distinguish** between self-pollination and cross-pollination
4. **Explain** how self-incompatibility works
5. **Outline** seed dispersal and germination
6. **Explain** how food reserves are mobilized

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between pollination and seed dispersal.
2. Give one way in which a hermaphrodite flower can avoid self-pollination.
3. Explain why oxygen is needed for germination.

Image credits

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Mature flower diagram.svg – Mariana Ruiz LadyofHats – Public domain
Misc pollen.jpg – Dartmouth College Electron Microscope Facility – Public domain
Holly.jpg – James E. Petts – CC BY-SA 2.0
Dandelion seeds.jpg – Author not stated – CC BY-SA 3.0
Coconut.jpg – Kulmalukko – CC BY 3.0
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